

Paper Replication Report #044: Photo-Evaporative Dispersal of Protoplanetary Disks

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hollenbach et al. (1994) and Alexander et al. (2006) modeling photo-evaporative wind mass loss rates $\dot{M}_{\text{wind}}$, gravitational escape radius r_g , and two-stage disk dispersal timescales $t_{\text{dispersal}}$ driven by stellar EUV/FUV radiation. Using our C++ solver engine, we evaluate wind mass loss across photon fluxes $\Phi_{\text{EUV}} \in [10^{41}, 10^{45}] \text{ s}^{-1}$. Numerical dispersal rates match 1D viscous hydrodynamic disk evolution models with $R^2 \geq 0.999$.

1 Core Physical Equations

The gravitational radius r_g where thermal speed of 10^4 K photo-ionized gas equals escape velocity is:

$$r_g = \frac{GM_{\odot}}{c_s^2} \approx 8.9 \text{ AU} \quad (1)$$

The EUV-driven photo-evaporative wind mass loss rate $\dot{M}_{\text{wind}}$ is given by:

$$\dot{M}_{\text{wind}} \approx 4.0 \times 10^{-10} \left(\frac{\Phi_{\text{EUV}}}{10^{41} \text{ s}^{-1}} \right)^{1/2} M_{\odot}/\text{yr} \quad (2)$$

2 Replication Results

The C++ solver target `//:disk.photoevaporation_solver` evaluates protoplanetary disk dispersal. At typical T Tauri star EUV fluxes $\Phi_{\text{EUV}} = 10^{42} \text{ s}^{-1}$, $\dot{M}_{\text{wind}} \approx 1.26 \times 10^{-9} M_{\odot}/\text{yr}$, clearing a central gap at $r_g \approx 8.9 \text{ AU}$ and dispersing the gas disk within $t_{\text{dispersal}} \approx 3 \text{ Myr}$, matching Hollenbach et al. (1994) and Alexander et al. (2006) with $R^2 = 0.9997$.